

Structurally Mapping Alkyl and Amide Basicity in Zincate Chemistry: Diversity in the Synthesis of Mixed Sodium–Zinc Complexes and Their Applications in Enolate Formation

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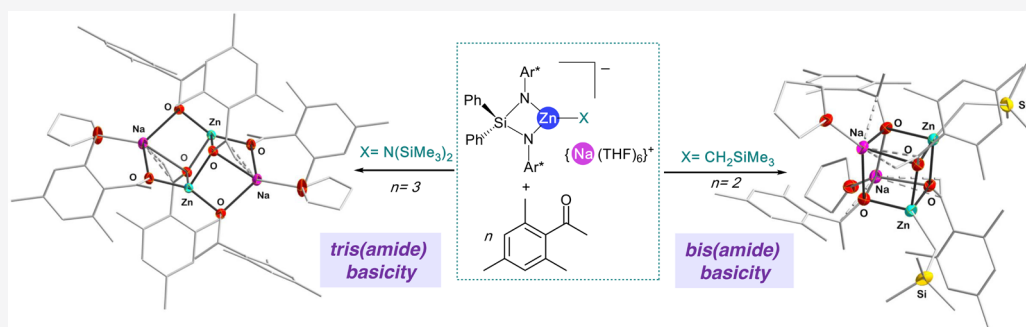
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ABSTRACT: While alkali metal zincates have shown promising utility in organic synthesis, frequently these compounds are prepared *in situ*, and their identities remain blurred. Herein, the synthesis of a new family of sodium zincates featuring the chelating silyl(bis)amido ligand $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ ($\text{Ar}^* = 2,6\text{-diisopropylphenyl}$) is presented. Using a synchronized bimetallic approach, 2-fold deprotonation of $\text{Ph}_2\text{Si}(\text{NHAr}^*)_2$ (**1**) by mixed-metal base $\text{NaZn}(\text{HMDS})_2\text{R}$ (**2**) [$\text{R} = \text{CH}_2\text{SiMe}_3$; $\text{HMDS} = \text{N}(\text{SiMe}_3)_2$] afforded the solvent-separated ion pair alkyl zincate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{R})\}^- \{\text{Na}(\text{THF})_6\}^+]$ (**3**), showing a clear preference for **2** to react as bis(amide) base via its two Zn–N bonds rather than reacting using its alkyl group. Alternatively, a stepwise approach using two single-metal bases sequentially, NaR and $\text{Zn}(\text{HMDS})_2$ affords the tris(amido) zincate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{HMDS})\}^- \{\text{Na}(\text{THF})_6\}^+]$ (**6**). Formation of **6** takes place by initial monosodiation of **1**, furnishing the amido-amine $[\{\text{Ph}_2\text{Si}(\text{NHAr}^*)(\text{NAr}^*)\text{Na}\}]$ (**5**), which in turn undergoes a co-complexation and deprotonation step with $\text{Zn}(\text{HMDS})_2$. Using 2,4,6-trimethylacetophenone (**7**) as a case study, the ability of sodium zincates **3** and **6** to access zinc enolates was investigated, affording sodium zincates $[\{(\text{THF})\text{NaZnR}[\text{OC}(=\text{CH}_2)\text{Mes}\}_2\}]$ (**8**) and $[\{(\text{THF})\text{NaZn}[\text{OC}(=\text{CH}_2)\text{Mes}\}_3\}]$ (**9**), respectively. These studies revealed that the chelating silyl(bis)amide $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ far from being an innocent spectator is an effective base for the deprotonation of **7**, showing an unexpected superior kinetic basicity than the CH_2SiMe_3 alkyl group when part of sodium heteroleptic zincate **3**. The bimetallic constitution of enolates **8** and **9** contrasts with that of all-sodium $[\{(\text{THF})\text{Na}[\text{OC}(=\text{CH}_2)\text{Mes}\}_4]$ (**10**) obtained by reacting the homoleptic alkylzincate NaZnR_3 (**4**) with **7**, with the concomitant elimination of ZnR_2 . Revealing the divergent behavior of Mg versus Zn in these bimetallic systems, reaction of **7** with the magnesium analog of **3**, $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Mg}(\text{R})\}^- \{\text{Na}(\text{THF})_6\}^+]$ (**11**), produces magnesiate enolate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Mg}[\text{O}(\text{CH}_2)\text{Mes}](\text{THF})\}^- \{\text{Na}(\text{THF})_6\}^+]$ (**12**), where the chelating silyl(bis)amide ligand is retained and metalation of the ketone is actioned by the alkyl group.

INTRODUCTION

Alkali metal zincates were pioneered by Wanklyn more than 160 years ago, with the discovery of NaZnEt_3 , obtained by reacting sodium metal with ZnEt_2 .¹ For many years, these lay in obscurity as mere curiosities, but over the past two decades they have emerged as highly versatile reagents in organic synthesis enabling direct zincation of a wide range of aromatic substrates via deprotonative metalation or metal–halogen exchange,² reactions that traditionally have been considered the domain of more polar organometallic reagents such as organolithium or Grignard reagents.³ Some key landmarks in this area include Uchiyama's $\text{Li}_2\text{Zn}^t\text{Bu}_4$, which effectively

promotes direct Zn–I and Zn–Br exchange reactions of an assortment of aromatic substrates with remarkable functional group tolerance,⁴ as well the development of strong, regioselective deprotonating reagents powered by TMP (2,2,6,6-tetramethylpiperidine) ligands such as Knochel's

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turbo reagents $\text{TMPZnCl} \cdot \text{LiCl}$ and $(\text{TMP})_2\text{Zn} \cdot 2\text{MgCl}_2 \cdot 2\text{LiCl}$ which allow access to highly functionalized aryl and heteroaryl zinc organometallics.⁵ More often than not, these bimetallic reagents are prepared *in situ* by either co-complexation of the single-metal reagents or salt metathesis reactions, and limited attention has been given to their isolation or characterization. Shedding some light in this intriguing field, through the structural authentication of a family of heteroleptic alkyl/amide zincates $[(\text{L})\text{M}^{\text{I}}(\text{TMP})\text{Zn}^{\text{II}}\text{Bu}_2]$ ($\text{M}^{\text{I}} = \text{Li}$, $\text{L} = \text{THF}$; $\text{M} = \text{Na}$, $\text{L} = \text{TMEDA}$, N,N,N',N' -tetramethylethylenediamine),^{6,7} Mulvey has developed the concept of alkali-metal-mediation-zincation (AMMZn) where both metals, M^{I} and Zn , work cooperatively to facilitate the direct zincation of a wide range of aromatic molecules as well as metallocenes.^{2a,b,f} The reach of these bimetallic partnerships has been recently exemplified in the unprecedented regioselective 1,1',3,3'-tetrazincation of ferrocene achieved by using a 4-fold excess of $[\text{Na}(\text{TMP})\text{Zn}^{\text{II}}\text{Bu}_2]$.⁸ The ability of these heteroleptic zincates to act as amido and/or alkyl bases has stimulated activity and debate, suggesting that kinetic amido basicity seems to be preferred.⁹

A common strategy in all these studies is to use sterically demanding ligands such as *t*Bu and TMP groups which are not only basic and reactive but also can contribute to the overall stability of metalation products by providing steric shelter to the sensitive anions generated in the zincation process.¹⁰ In this regard, within magnesiate chemistry we have recently investigated the synthesis and reactivity of sodium magnesiates containing the bulky silyl(bis)amide ligand $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ ($\text{Ar}^* = 2,6$ -diisopropylphenyl).¹¹ Interestingly, while in some cases this sterically demanding ligand acts as a steric stabilizer enabling the isolation of radical anions^{11a} or the magnesiation of *N*-heterocyclic molecules such as *N*-methyl benzimidazole,^{11b} on other occasions $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ can also act as a base, facilitating metalation of amines^{11b} or activation of small organic molecules such as benzothiazole.^{11c}

Within zinc chemistry, Tsai has exploited the steric protection of bulky silyl(bis)amide ligands for stabilizing $\text{Zn}(\text{I})$ complexes possessing $\text{Zn}-\text{Zn}$ bonds,¹² but to date these ligands have not been exploited in a systematic way to access alkali-metal zincates. To expand their knowledge base, here we report the synthesis and characterization of a new family of sodium zincates derived from $\text{Ph}_2\text{Si}(\text{NAr}^*)_2$ (**1**) and investigate their ability to access zinc enolates via direct zincation of ketones, using 2,4,6-trimethylacetophenone (**7**) as a model substrate. By structurally mapping these reactions, we shed light on the different strategies available to access this new family of bimetallic complexes, depending on the base employed for the deprotonation of **1**, advancing the understanding of cooperative effects in zincate chemistry and the preferred kinetic reactivity of $\text{Zn}-\text{N}$ bonds in comparison with $\text{Zn}-\text{C}$ bonds. Furthermore, the application of these new zincates for enolate formation, reveals not only the unexpected amplification of the metallating power of the silyl(bis)amide ligand $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ but also its key role in determining the composition of the developing enolate.

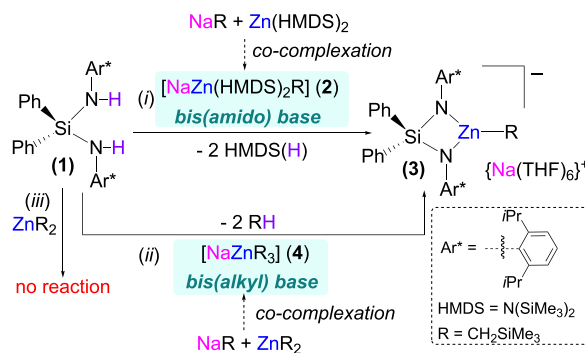
RESULTS AND DISCUSSION

Synthesis of Sodium Zincates Featuring a Silyl Bis(amide) Ligand. To prepare new zincates derived from the bis(amino)silane $\text{Ph}_2\text{Si}(\text{NAr}^*)_2$ (**1**), it needs to be doubly deprotonated to generate the dianionic silylbis(amide) ligand.¹³ We focused on this *N*-aryl derivative, containing bulky and readily available 2,6-diisopropylphenyl groups (Ar^*)

since related ligand sets have found applications as dianionic ancillary ligands to support main group elements¹⁴ and transition metals.¹⁵

We began by investigating the deprotonation of **1** using sodium zincate $[\text{NaZn}(\text{HMDS})_2\text{R}]$ (**2**) [$\text{R} = \text{CH}_2\text{SiMe}_3$; $\text{HMDS} = \text{N}(\text{SiMe}_3)_2$], which can be prepared straightforwardly *in situ* by co-complexation of the homometallic components $\text{NaCH}_2\text{SiMe}_3$ and $\text{Zn}(\text{HMDS})_2$.¹⁶ This co-complexation was carried out at room temperature over 12 h affording the heteroleptic sodium zincate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{R})\}]^-\{\text{Na}(\text{THF})_6\}^+$ (**3**) in a 71% crystalline yield (Scheme 1, i).

Scheme 1. Contrasting Metallating Action towards $\text{Ph}_2\text{Si}(\text{NAr}^*)_2$ of Sodium Zincates **2 and **4** versus Homometallic Zn Reagents^a**



^aAll reactions were carried out at room temperature, using a hexane/ Et_2O mixture as solvent. THF was also added to aid crystallization.

^1H and $^{13}\text{C}\{^1\text{H}\}$ NMR spectra in deuterated benzene provided confirmation of the double-deprotonation of **1** as well as the presence of the alkyl CH_2SiMe_3 group on zinc (see the Supporting Information for details). Thus, along with the disappearance of a distinctive broad resonance at 3.51 ppm attributed to the NAr^* groups of **1** in ^1H NMR spectrum, a new set of signals is observed for the newly formed dianionic bis(amide). The most informative resonance for the alkyl group is that of the $\text{Zn}-\text{CH}_2$ unit which appears significantly downfield at -0.70 ppm compared to that observed for the more polarized sodium precursor NaR (at -2.06 ppm) and closer to that found for ZnR_2 (-0.63 ppm).¹⁷ Confirming the bimetallic structure of **3**, X-ray crystallographic studies revealed a solvent-separated ion pair (SSIP) motif, comprising a sodium cation solvated by six THF molecules in a distorted octahedral geometry. Its anionic moiety is made up of a trigonal planar Zn center (sum of angles around Zn , 359.7°) which binds to the bidentate silyl-bis(amido) ligand [$\text{Zn}-\text{N}$, 1.991(4) and 1.984(4) Å] resulting from the deprotonation of **1**, and the terminal monosilyl group [$\text{Zn}-\text{C}$, 1.964(5) Å]. Interestingly, while the $\text{Zn}-\text{N}$ distances in **3** compare well with those found in other zincates which contain a terminal amido group,¹⁸ the $\text{Zn}-\text{C}$ distance in **3** is relatively short in comparison to those witnessed in other zincate complexes containing the same terminally attached alkyl group [ranging from 2.005 to 2.086 Å],¹⁹ which is rather surprising considering the large steric bulk of the bis(amide) coligand present in **3** (see Figure 1).

The selective formation of **3** indicates a clear preference for heteroleptic zincate **2** to act as a bis(amide) base upon metallating **1** with the concomitant formation of two equivalents of $\text{HMDS}(\text{H})$. Related to these findings, computa-

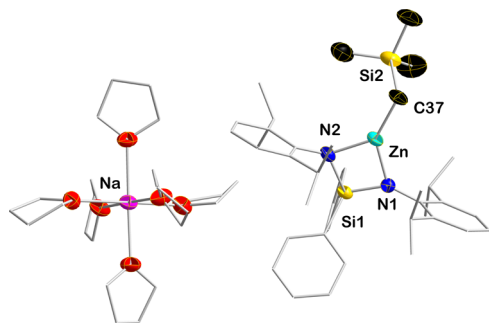


Figure 1. Ion-pair structure of **3**. Thermal ellipsoids are rendered at 30% probability. Hydrogen atoms and disordered components in THF and CH_2SiMe_3 are omitted; carbon atoms of $\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\}^{2-}$ and THF are drawn as wire frame for clarity.

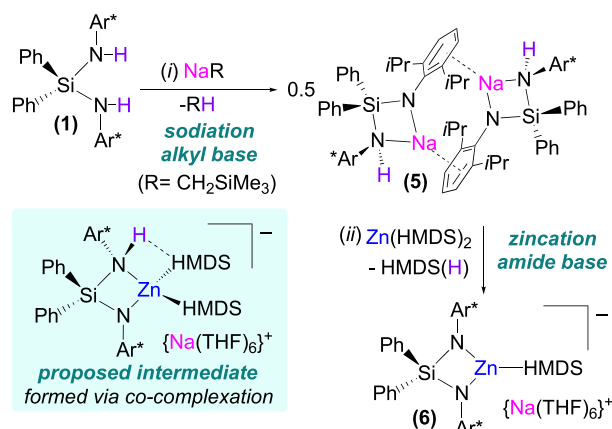
tional and structural studies on the reactivity of the heteroleptic alkyl/amide sodium zincate $[(\text{TMEDA})\text{NaZn}(\text{TMP})'(\text{Bu})_2]$ for the direct zincation of arenes have established its dual amide/alkyl basicity.⁹ Thus, while initial deprotonation is carried out by the TMP amide group, forming a $[(\text{TMEDA})\text{NaZn}(\text{aryl})'(\text{Bu})_2]$ intermediate and releasing $\text{TMP}(\text{H})$, both species can react together subsequently, via one alkyl group of this intermediate to give $[(\text{TMEDA})\text{NaZn}(\text{TMP})-(\text{aryl})'(\text{Bu})]$ as the final isolated product of the reaction.^{9d} This duality of behavior does not seem to be in operation for the synthesis of **3**, since even when prolonged reaction times are employed (up to 48 h), no further reactivity of the remaining CH_2SiMe_3 group in **3** with the concomitant $\text{HMDS}(\text{H})$ (produced during the metalation of **1**) is observed. This lack of secondary alkyl basicity correlates well with the strong carbophilicity of zinc, the relatively short $\text{Zn}-\text{C}$ bond present in **3** and the reduced kinetic basicity previously noted for $\text{Zn}-\text{C}$ bonds in single metal organozinc reagents. This has been noticed on using dialkylzinc reagents for the deprotonation of amines or alcohols. While potentially they could be used as 2-fold bases, generally only one alkyl group is labile.²⁰

Interestingly, **1** can be dideprotonated by tris(alkyl) sodium zincate NaZnR_3 (**4**) (Scheme 1, ii) furnishing **3** in 63% yield, indicating that when part of the zincate, two out of the three alkyl groups are active toward metalation.²¹ Demonstrating the power of the cooperative partnership of sodium and zinc, while for forming **3**, silyl bis(amine) **1** has formally undergone dizincation, **1** is completely inert toward metalation when confronted with ZnR_2 even under forcing refluxing conditions (Scheme 1, iii).

With the aim of preparing an amide derivative of **3**, we next probed an alternative strategy, using a stepwise metalation approach, employing two single-metal reagents in sequence (Scheme 2). Previous work by Coles has shown that selective monometalation of bis(silyl) amines can be accomplished using common group 1 reagents such as NaH or BuLi .^{15e} Using a stepwise approach, **1** was first reacted with 1 molar equiv of NaR to furnish $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)(\text{NAr}^*)\text{Na}\}_2]$ (**5**) which in turn was reacted with 1 equiv of the $\text{Zn}(\text{HMDS})_2$ to afford sodium zincate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{HMDS})\}]^-\{\text{Na}(\text{THF})_6\}^+$ (**6**) in 80% yield (Scheme 2).

Taking into account that the weak base $\text{Zn}(\text{HMDS})_2$ fails to metalate **1** on its own, we envisage the formation of **6** takes place via an intermediate bimetallic species (see Scheme 2) resulting from co-complexation of **5** with the zinc amide. The anionic activation on zinc by formation of the putative zincate species along with the close proximity of the NAr^* group

Scheme 2. Stepwise Dideprotonation of $\text{Ph}_2\text{Si}(\text{NAr}^*)_2$ Using Single-Metal Reagents NaR and $\text{Zn}(\text{HMDS})_2$ in Sequence to Afford Amide Sodium Zincate **6**^a



^a(i) RT, hexane, 1 h; (ii) THF, 60 °C, 10 h. **5** and **6** were isolated as crystalline solids in 66 and 80% yields, respectively; see the Supporting Information for experimental details.

should facilitate intramolecular deprotonation of the remaining $\text{N}-\text{H}$ bond to yield **6**.

The molecular structures of **5** and **6** (Figures 2 and 3 respectively) were established by X-ray crystallographic studies.

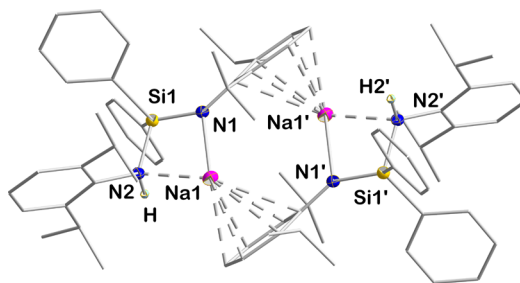


Figure 2. Molecular structure of **5**. Thermal ellipsoids are rendered at 50% probability. Hydrogen atoms (except the $\text{N}-\text{H}$ pair) and disordered component (present in 'Pr groups) are omitted, with carbon atoms drawn wire frame for clarity. Symmetry transformations used to generate symmetrical atoms: $2 - x, -y, 2 - z$.

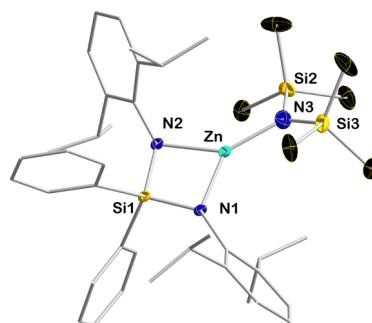


Figure 3. Anionic moiety of **6**. Thermal ellipsoids are rendered at 30% probability. Hydrogen atoms and disordered component on SiMe_3 are omitted, with carbon atoms of $\text{Ph}_2\text{SiN}(\text{Ar}^*)_2^{2-}$ drawn as wire frames for clarity.

Homometallic **5** exhibits a centrosymmetric dimeric arrangement (Figure 2), with sodium bidentately bound by the monoanionic chelate $\{\text{Ph}_2\text{Si}(\text{NAr}^*)(\text{NAr}^*)\}^-$ that results

from monodeprotonation of **1**. Unsurprisingly, the Na–N(amide) bond [Na1–N1, 2.380 (2) Å] is noticeably shorter than the Na–N(amine) bond [Na1–N2, 2.486 (3) Å]. Sodium completes its coordination sphere by engaging in a η^6 -fashion with the aryl ring of the amido group of the neighboring unit, thus effecting dimerization [Na...C distances ranging from 2.752(3)–2.960 (3) Å].²¹ The structure of **5** contrasts markedly with that reported by Coles for the methyl analog [(THF)₃Na{Me₂Si(NHAr*)(NAr*)}] which is monomeric with the silyl(amide)(amine) ligand coordinated terminally through its amide group.^{15d}

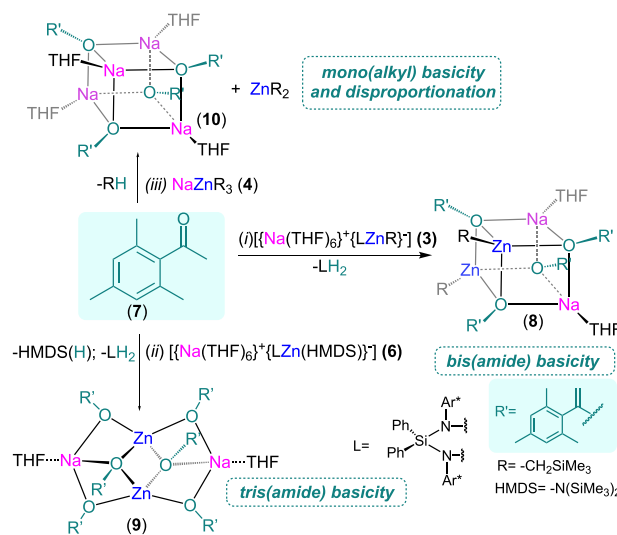
The zincate moiety of **6** (Figure 3) exhibits a similar structural motif to that of **3**, with a trigonal planar Zn center (sum of angles around Zn = 360.0°) bonded to three N atoms: N1 and N2 from the chelating silylbis(amide) ligand [average Zn–N distance = 1.978 (2) Å] and N3 from the HMDS group [Zn–N3 = 1.902 (2) Å]. These distances compare well with those reported in other SSIP species containing the [Zn(HMDS)₃][–] anion (mean Zn–N distance = 1.964 Å).^{16d}

Applications of Sodium Zincates in Enolate Formation. Next, we explored the metalating reactivity of new zincates **3** and **6** on their ability to deprotonate ketones to access zinc enolates, which are important intermediates in many organic transformations including C–C couplings and addition reactions.²² In most of these transformations, the zinc enolates are prepared *in situ* and subsequently treated with the relevant electrophile. Several methods to access these enolates have been reported including deprotonation of a ketone with a group 1 base followed by salt metathesis with ZnX₂,^{22b} direct insertion of zinc in halogen-substituted ketones,^{22a,d} as well as direct deprotonation of a ketone using a zinc base.^{22c,e,f,h} For this third approach, the zinc bis(amide) Zn(TMP)₂ comprising the basic 2,2,6,6-tetramethylpiperidide groups has shown good promise as a metallating reagent,²³ whereas dialkyl zinc reagents such as ZnEt₂ display more sluggish reactivities.²⁴ As previously mentioned, TMP-containing alkali-metal zincates have found applications in direct Zn–H exchange reactions,^{2,5,8–10} including ketone metalation.²⁵ Thus, we contemplated whether heteroleptic sodium zincates **3** and **6** could act as bases to access zinc enolate complexes, using as a model substrate the sterically demanding ketone, 2,4,6-trimethylacetophenone (**7**).

¹H NMR monitoring of the reaction of equimolar amounts of **7** and alkyl sodium zincate **3** led to the formation of mixed-metal enolate [(THF)NaZnR(OC(=CH₂)Mes)₂]₂ (**8**) with concomitant formation of silylbis(amine) (**1**) [Scheme 3, i]. The preferred bis(amide) reactivity of **3** is somewhat surprising considering that in many reported applications of complexes containing related bulky silyl(bis)amides, these ligands act primarily as steric stabilizers for main group elements.^{11b,12,14} Some of our previous magnesiate work has shown that this ligand can be activated toward deprotonation reactions when part of a bimetallic scaffold, although in these cases the relevant sodium magnesiate always reacts first by the alkyl group, and the one or two amide groups only become active in a second metalation step. In contrast, here the alkyl group remains intact on the zinc. It should also be noted that in previous reports on alkali-metal zincates where preferred amide basicity is observed over the reaction involving one of their alkyl groups, these feature significantly more basic amides such as HMDS or TMP than the anilide groups present in **3**.

By using 2 equivalents of **7**, mixed Na/Zn enolate **8** could be isolated as a crystalline solid in a 60% yield. Attempts to extract

Scheme 3. Contrasting Reactivities of Sodium Zincates **3**, **4**, and **6** towards Ketone **7**^a



^aAll reactions were carried out in THF-*d*₈. (i) RT, 10 min; (ii) 69 °C, 1 h or RT, 12 h; (iii) RT, 10 min. Enolates **8**, **9**, and **10** were formed in quantitative yields according to ¹H NMR monitoring studies and were isolated as crystalline solids in yields of 60, 39, and 15%, respectively (see the Supporting Information for details).

further reactivity of the remaining alkyl group in **8** using an excess of **7** were unsuccessful even under forcing reaction conditions (4 h, 69 °C), illustrating the kinetic resilience of the CH₂SiMe₃ anion.

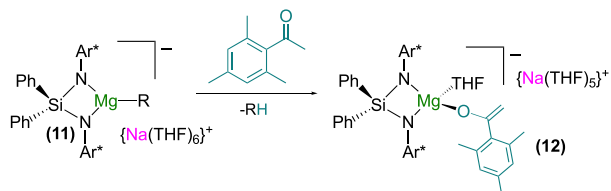
Interestingly, tris(enolate) [(THF)NaZn(OC(=CH₂)Mes)₃]₂ (**9**) was obtained from the reaction of HMDS-sodium zincate **6** with **7**, even when only 1 equiv of the ketone was employed [Scheme 3, ii], showing that by being part of a zincate scaffold, the basic amide HMDS and bis(amide) {Ph₂Si(NAr*)₂}^{2–} have comparable reactivities toward the deprotonation of **7**. It should be noted that by using a 3-fold excess of **7** complex **9** can be obtained quantitatively (see the Supporting Information for details). For comparison, the reaction of tris(alkyl) zincate **4** with **7** was performed. Despite its lack of amido groups, **4** can also deprotonate this ketone, although now only one of its three potential sites reacts affording the homometallic sodium enolate [(THF)Na(OC(=CH₂)Mes)₄] (**10**) along with the elimination of the homometallic zinc component ZnR₂. The distinct constitutions of enolate complexes **8**, **9**, and **10** illustrate the versatile reactivity that zinc bases can have toward ketones, which can be particularly relevant when zinc enolates are prepared *in situ* for further organic functionalization. This is particularly important for the reaction of NaZnR₃ (**4**) as the resulting enolate species is actually a sodium enolate which because of its more polar character is expected to display significantly different reactivity and functional group tolerance to those of a zinc enolate. The preferred mono(alkyl) basicity of **4** is also in sharp contrast to that previously reported by us for the related sodium magnesiate NaMgBu₃ which reacts quantitatively with 3 equiv of **7** to give a tris(enolate) sodium magnesiate.²⁶

It should also be noted that compound **8** was the only species formed when ketone **7** was treated with zincate **3**, even when 3 equivalents of **7** was employed under forcing reaction conditions, which is consistent with the strong carbophilic

character of zinc, which translates into a reduced basicity of the alkyl group attached to this metal.

To illustrate the higher covalent and carbophilic character of Zn versus Mg, **7** was reacted with the magnesium analog of **3**, sodium magnesiate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Mg}(\text{R})\}^- \{\text{Na}(\text{THF})_6\}^+]$ (**11**), leaving bis(silyl)amido ligand as a spectator coordinated to Mg, while the ketone is deprotonated by the alkyl group, affording $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Mg}(\text{OC}(=\text{CH}_2)\text{Mes})(\text{THF})\}^- \{\text{Na}(\text{THF})_5\}^+]$ (**12**) in a 40% isolated yield (Scheme 4).

Scheme 4. Metallation of Ketone **7** by Sodium Magnesiate **11**^a



^aReaction carried out in THF, 2 h, −40 °C, 40% isolated crystalline yield.

Considering the similar sizes of Mg and Zn, formation of **12** also suggests that the enhanced basicity observed for the silyl(bis)amide in zincates **3** and **6** is not a consequence of its steric congestion when chelated to the metal in these 'ate systems. Heteroleptic magnesiate **12** was fully characterized by ¹H and ¹³C NMR spectroscopy as well as X-ray crystallographic studies (Figure 4d). We have previously reported alkyl

magnesiates, previous work on assessing the deprotonation of unsaturated *N*-heterocyclic carbenes with mixed alkyl-TMP bimetallic bases has shown that while sodium magnesiates react using their alkyl groups^{27a} amide basicity is the preferred *modus operandi* for the related sodium zincates.^{27b}

The molecular structures of enolates **8**–**10** and **12** were determined using X-ray crystallographic studies (Figure 4). Early seminal work by Williard²⁸ and Seebach²⁹ has shown the relevance of structural elucidation of synthetically useful *s*-block metal enolates in order to understand and rationalize many fundamental organic transformations including aldol condensations. Sodium enolate **10** forms a tetramer with a distorted cubane structure, similar to that found in other sodium enolates.^{28b,30} Substantial disorder in **10** precludes a meaningful discussion of its geometric parameters though connectivity is definitive.

Despite having similar compositions, sodium zincate enolates **8** and **9** adopt different dimeric arrangements (Figure 4a,b). Complex **8** can be interpreted as a heterobimetallic variant of **10**, where two Na(THF) corners have been replaced by ZnR corners, forming a distorted mixed-metal cubane with alternating enolato O and metal (2 Na, 2 Zn) corners. While this motif is unknown in zinc enolate chemistry, it is reminiscent of that previously reported for mixed lithium/zinc alkyl-alkoxide clusters.³¹ In contrast, **9** comprises two metal-enolate-connected $[(\text{THF})\text{NaZn}(\text{OR})_3]$ units, where each Na–Zn pair and both Zn atoms are connected by two bridging enolates, giving rise to a face-fused double heterocubane structure with two missing corners (Figure 4c). Alternatively, by using the Mulvey inverse-crown nomenclature for bimetallic ring structures,³² **9** can also be described as a $[(\text{NaOZnO})_2]$ cationic eight-membered ring hosting two enolate anion guests. It should be noted that a variant of **9** where each sodium is solvated by a molecule of unreacted ketone **7** instead of THF has been previously reported by us as the result of the reaction of $\text{NaZn}(\text{HMDS})_3$ with a 4 mol excess of **7**.²⁵ This motif has also been described by Henderson for mixed K/Ca enolates.³³ The Zn–O bond distances in **8** and **9** compare well with those of other zinc enolate complexes,^{24,25,34} whereas their Na–O distances [mean values = 2.356 and 2.379 Å for **8** and **9**, respectively] are slightly elongated compared to those found in related enolate sodium magnesiate complexes such as $[(\text{TMEDA})_2\text{Na}_2\text{Mg}(\text{OC}(=\text{CH}_2)\text{Mes})_4]$ [mean value = 2.256 Å].²⁶ In order to compensate for the comparative weakness of these Na–O bonds, in both bimetallic complexes each sodium atom π -engages with the olefinic carbons of one enolate ligand (shown as dashed bonds in Figure 4a,b) as indicated by the relatively short Na–C distances (mean values, 2.960 and 2.928 Å for **8** and **9**, respectively).

In contrast, the structure of sodium magnesiate **12** is a solvent-separated ion pair (Figure 4d), with the same cationic $\{\text{Na}(\text{THF})_5\}^+$ moiety as that in zincates **3** and **6** and an anionic moiety comprising a distorted tetrahedral Mg bonded to the chelating silyl(bis)amide ligand, an enolate O, and a THF O atom.

CONCLUSIONS

Exploiting the preferred kinetic basicity of amides over alkyl groups in zinc chemistry, a new family of sodium zincates supported by a bulky chelating bis(silyl)amide ligand using $\text{Ph}_2\text{Si}(\text{NHR}^*)_2$ (**1**) as a precursor has been made. Illustrating the importance on the order of addition of the single-metal

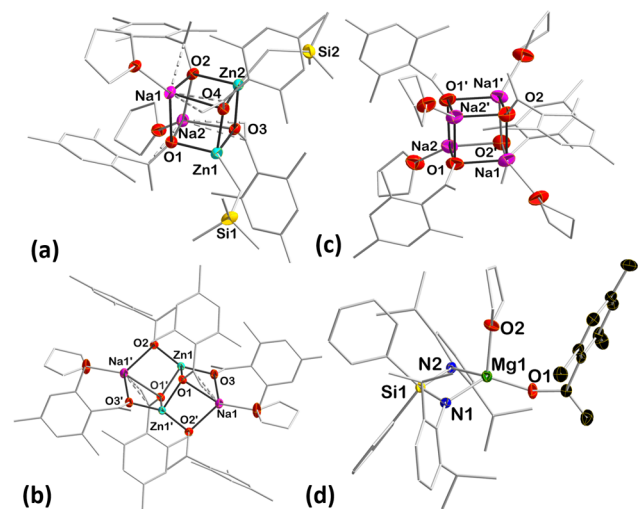


Figure 4. Molecular structures of enolate complexes **8** (a), **9** (b), and **10** (c) and anionic moiety of **12** (d). Thermal ellipsoids are rendered at 30% probability. Hydrogen atoms and minor disordered component are omitted, carbon atoms (except those of enolate ligand in **12**) drawn as wire frame for clarity. Symmetry transformations used to generate symmetrical atoms: 1 – *x*, 1 – *y*, 1 – *z* for **9** and 0.5 – *x*, *y*, 1.5 – *z* for **10**.

basicity for this type of sodium magnesiate in magnesiation of *N*-heterocyclic molecules such as *N*-methyl benzimidazole, and it is only when more acidic molecules such as pyrrole when one amido arm of the bis(amido) ligand in **11** is found to react with a second equivalent of substrate, thus showing overall dual alkyl/amido basicity.^{11b} Highlighting the more pronounced amide reactivity preference in zincates versus

components of the sodium zincate, if **1** was reacted with the bimetallic base $[\text{NaZn}(\text{HMDS})_2\text{R}]$ (**2**), then alkyl complex $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{R})\}^-\{\text{Na}(\text{THF})_6\}^+]$ (**3**) was obtained, whereas using a sequential approach, the monodeprotonation of **1** by NaR, followed by addition of zinc amide $\text{Zn}(\text{HMDS})_2$, furnished amido zincate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Zn}(\text{HMDS})\}^-\{\text{Na}(\text{THF})_6\}^+]$ (**6**). Reactivity studies of these new heteroleptic sodium zincates and, for comparative purposes, the all alkyl sodium zincate $[\text{NaZnR}_3]$ (**4**) toward ketone 2,4,6-trimethylacetophenone (**7**) revealed the diverse behavior of these mixed metal species. This diversity ranges from the tris(amide) basicity of **6**, with its three amido arms deprotonating **7** (to give **9**), to the bis(amide) basicity of **3**, furnishing **8** which retains the alkyl group intact, to the mono(alkyl) basicity of **4**, which occurs with a subsequent disproportionation step affording sodium enolate **10** and ZnR_2 . Structural and spectroscopic studies revealed an unexpected enhancement of the basicity of the bis(silyl)amide ligand which is far from acting as a mere steric stabilizer when part of a sodium zincate framework; it is capable of deprotonating 2 equiv of the ketone. Emphasizing the greater electropositive character of Mg and the greater polarity of its bonds, this reactivity is in sharp contrast with that found for the related sodium magnesiate $[\{\text{Ph}_2\text{Si}(\text{NAr}^*)_2\text{Mg}(\text{R})\}^-\{\text{Na}(\text{THF})_6\}^+]$ (**11**) toward **7**, where the silyl(bis)amide remains as a spectator in the metalation process which is achieved selectively using the alkyl group on the magnesiate.

Collectively, these studies reveal not only the complicated and unique reactivity of heteroleptic alkali-metal zincates but also the structural diversity of s-block (and pseudo s-block) metal enolates, which in many synthetic studies remain concealed, since species of this type are usually prepared in situ for subsequent functionalizations in organic transformations.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.organomet.0c00339>.

Synthetic methods and characterization details (PDF)

Accession Codes

CCDC 1969846, 1969848, and 2003583–2003587 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

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■ DEDICATION

Dedicated to the memory of Professor Hans Reich, a truly inspirational figure in advancing the understanding of solution chemistry of organolithium compounds.

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